

Chapter 12: Comparison Method and Stability of EPCA.

10

Goal:

Systems of nonlinear differential equations with piecewise constant arguments (EPCAs) $\Rightarrow$ $\left\{ \begin{array}{l} \text{comparison principle.} \\ \Downarrow \\ \text{stability properties.} \end{array} \right.$

Results: piecewise (constant) arguments
un stable $\xrightarrow{\hspace{10em}}$ stable.

Introduction: nonlinear EPCA has the form

$$\dot{x} = f(t, x(t), x(r(t))).$$

where the argument r is a piecewise constant function defined on intervals with a certain length. For example,

$$r(t) = [t], [t-n], t-n[t], [t+1]. \quad \forall t, n \in \mathbb{N}. \quad [\cdot] \text{ integer function}$$

problem Formulation.

Let $\{t_k\}_{k=0}^{\infty}$ and $\{\xi_k\}_{k=0}^{\infty}$ be sequences of nonnegative real numbers such that $\lim_{k \rightarrow \infty} t_k = \infty$, $t_{k+1} \leq \xi_k \leq t_k$, $\xi_0 = t_0$.

Consider the EPCA of the form.

$$\dot{x}(t) = f(t, x(t), \lambda_{p(t)}(x(r(t))). \quad (12.2a).$$

$$p(t) = k, \quad r(t) = \xi_k, \quad t \in [t_k, t_{k+1}] \quad \text{C piecewise constant functions}$$

$$x(t_0) = x_0.$$

$$(12.2b).$$

For (12.2), we state the solution of IVP.

Definition 12.1. $x: (a, b) \rightarrow \mathbb{R}$ is said to be a solution of (12.2)

if the following conditions hold:

- (i) $x(t)$ is continuous for all $t \in (a, b)$;
- (ii) $\dot{x}(t)$ exists and continuous for all $t \in (a, b) \setminus \{t_k\}$;
- (iii) $\dot{x}(t)$ satisfies the EPCA in (12.2a);
- (iv) $x(t)$ satisfies the initial condition in (12.2b) at $t = t_0$.

We can rewrite (12.2) as follows.

$$\begin{cases} \dot{x}(t) = f(t, x(t), \lambda_k(x_{s_k})), & t \in [t_k, t_{k+1}), \quad k=0,1,2,\dots \quad (12.3a) \\ x(t_0) = x_0 \end{cases} \quad (12.3b)$$

We employ the theory of ODE to show the solution of (12.3).

For $k=0$, $t \in [t_0, t_1)$, the IVP

$$\begin{cases} \dot{x}(t) = f(t, x(t), \lambda_0(x_{s_0})), \\ x(t_0) = x_0, \quad s_0 = t_0 \end{cases}$$

has a unique solution, say $x_0(t)$, and $x_0(t_1) = \lim_{t \rightarrow t_1^-} x_0(t)$.

Similarly, for $k=1$ and $t \in [t_1, t_2)$, we have the IVP.

$$\begin{cases} \dot{x}(t) = f(t, x(t), \lambda_1(x_{s_1})), \\ x(t_1) = x(t_1^-), \end{cases}$$

which has a unique solution, say $x_1(t)$.

By induction, we have

$$x(t) = \begin{cases} x_0 & t = t_0 \\ x_0(t, t_0, x_0) & t \in [t_0, t_1) \\ x_1(t, t_1, x_1) & t \in [t_1, t_2) \quad x_1 = x_0(t_0^-, t_0, x_0) \\ \vdots \\ x_k(t, t_k, x_k) & t \in [t_k, t_{k+1}) \quad x_k = x_{k-1}(t_{k-1}^-, t_{k-1}, x_{k-1}). \end{cases}$$

To further analyze the comparison principle, we define auxiliary ^{scalar} system

$$\begin{cases} \dot{u}(t) = g(t, u(t), b_k(u_{s_k})) \end{cases} \quad (12.4a)$$

$$u(t_0) = u_0 \quad (12.4b)$$

Definition 12.2. (Lyapunov Functional).

If $V \in \mathcal{V}([t_k, t_{k+1}) \times \mathbb{R}^n, (\mathbb{R}_+))$, the upper right-hand derivative of

V is defined by

$$D^+ V(t, x, y) = \limsup_{h \rightarrow 0^+} \frac{1}{h} [V(t, x + h f(t, x, \lambda_k(y))) - V(t, x)].$$

Moreover, if $V \in \mathcal{V}([t_k, t_{k+1}) \times \mathbb{R}^n, (\mathbb{R}_+))$, then

$$D^+ V(t, x, y) = \frac{\partial V(t, x)}{\partial x} \cdot f(t, x, \lambda_k(y)).$$

12.3. Comparison Method.

Theorem 12.1. Assume that the following conditions hold: ③.

- (i) for $k=0,1,2, \dots$, $V \in \mathcal{V}([t_k, t_{k+1}) \times \mathbb{R}^n; \mathbb{R}^+)$, $V(t, x)$ is locally Lipschitz in x and.

$$D^+ V(t, x, V_{s_k}) \leq g(t, V(t, x), G_k(V_{s_k})), \quad t \in (t_k, t_{k+1})$$

where $V_{s_k} = V(s_k, x(s_k))$; and.

- (ii) the maximal solution $\Theta(t; t_0, u_0)$ of auxiliary scalar EPCA- (12.4) exists on $[t_0, \infty)$.

Then $V(t_0, x_0) \leq u_0$ implies $V(t, x(t)) \leq \Theta(t; t_0, u_0)$, $t \geq t_0$.

Proof: Define $m(t) = V(t, x(t))$.

$$D^+ m(t) \leq g(t, m(t), G_k(m_{s_k})), \quad t \in (t_k, t_{k+1}).$$

Thus, for $t \in [t_0, t_1]$, by classical comparison principle [6].

$$m(t) \leq \Theta_0(t; t_0, u_0), \quad t \in [t_0, t_1],$$

where $\Theta_0(t; t_0, u_0)$ is the maximal solution of auxiliary scalar.

$$\text{IVP} \quad \begin{cases} u_{i+1} = g(t, u_{i+1}, G_0(u_{s_i})). \\ u(t_0) = u_0 \end{cases}$$

By induction, we define

$$u_{i+1} = \begin{cases} u_0 & t = t_0 \\ \Theta_0 & t \in [t_0, t_1] \\ \Theta_1 & t \in [t_1, t_2] \\ \vdots & \vdots \end{cases}$$

Then, for $t \geq t_0$, we get.

$$m(t) \leq u(t) \Rightarrow m(t) \leq \Theta(t; t_0, u_0)$$

With the above preparations, we address some special cases of EPCA and EPCA η .

Corollary 12.1. Suppose that the condition in Theorem 12.1 hold,
 Let $k=0, 1, 2, \dots$ and $t \in [t_k, t_{k+1}]$. If we choose.

(i) $g(t, u, \phi_k(u_{s_k})) = \beta_k u_{s_k}$, with β_k being a constant for all k , then

(i) $s_k = t_k$, we have

$$V(t, x(t)) \leq \begin{cases} [1 + \beta(t-t_0)] V(t_0, x_0) & k=0, t \in [t_0, t_1] \\ [1 + \beta_k(t-t_k)] \prod_{j=1}^k [1 + \beta_{j-1}(t_j - t_{j-1})] V(t_0, x_0) & k \in \mathbb{N}, t \in [t_k, t_{k+1}] \end{cases}$$

where $t_k < t_{k+1}$ if $\beta_k > 0$ and $t_{k+1} < t_k - \frac{1}{\beta_k}$ if $\beta_k < 0$;
 (2) $t_{k+1} \leq t_k$, ~~where~~ we have

$$V(t, x(t)) = V_0(t, x(t)) \leq [1 + \beta_k(t-t_0)] V_0(t_0, x_0).$$

$\forall t \in [t_0, t_1]$, such that $t_1 - t_0 < -\frac{1}{\beta_0}$ and

$$V(t, x(t)) = V_k(t, x(t)) \leq V_{k-1}(t_k, x(t_k)) + \beta_k(t-t_k) V_{k-1}(s_k, x(s_k))$$

for any $t \in [t_k, t_{k+1}]$, $t_{k+1} - t_k < -\frac{c_k}{\beta_k c_{s_k}}$ where $c_k = V_{k-1}(t_{k+1}, x(t_{k+1})) / c_{s_k} = V_{k-1}(s_k, x(s_k))$

(ii) $g(t, u, \phi(u_{s_k})) = \alpha u(t) + \beta_k u_{s_k}$

(iii) $g(t, u, \phi(u_{s_k})) = \alpha u(t) + h(t, u, \phi_k(u_{s_k}))$.

proof. we only prove (i).

(i) (1) For $t \in [t_k, t_{k+1}]$, since $u_{s_k} = u_{t_k}$, the solution of the differential equation $\dot{u}(t) = \beta_k u_{s_k}$ is given by

$$u(t) = [1 + \beta_k(t-t_k)] u_k.$$

In particular, for $k=0, t \in [t_0, t_1]$, we have

$$u(t) = [1 + \beta_0(t-t_0)] u_0.$$

for $k=1, t \in [t_1, t_2]$, we have

$$u(t) = [1 + \beta_1(t-t_1)] [1 + \beta_0(t_1-t_0)] u_0.$$

Thus. by induction, we reach

$$u(t) = \begin{cases} [1 + \beta_0(t-t_0)] u_0, & k=0, t \in [t_0, t_1] \\ [1 + \beta_k(t-t_k)] \prod_{j=1}^k [1 + \beta_j(t_j - t_{j-1})] u_0, & k \in \mathbb{N}, t \in [t_k, t_{k+1}] \end{cases}$$

by comparison results developed in Th 2.1, we complete the proof.

(i) (2). For any $k, t \in [t_k, t_{k+1}]$

$$u(t) = u(t_k) + \beta_k(t-t_k) u(\xi_k).$$

For $k=0, \xi_0 = t_0$

$$u(t) = [1 + \beta_0(t-t_0)] u_0 =: u_0(t). \quad (\beta_0 < 0, t \in [t_0, t_1])$$

For $k=1, \xi_1, u(t) = u(t_1) + \beta_1(t-t_1) u_0(\xi_1) =: u_1(t)$

by induction, we reach.

$$u(t) = u_k(t) = u_{k-1}(t_k) + \beta_k(t-t_k) u_{k-1}(\xi_k).$$

This implies the desired estimates.

12.4. Stability Analysis.

Having established the comparison results in Theorem 12.1, we provide some stability notions for the nonlinear EPCA.

Theorem 12.2. In addition to the conditions in Theorem 12.1, assume there exist class $-K$ function a and b such that

$$b(\|x\|) \leq V(t, x) \leq a(\|x\|) \quad (12.6)$$

Then stability of $u=0$ in (12.4) $\Rightarrow$ stability of $x=0$ in (12.3).

proof. Fix $t_0 \in \mathbb{R}_+, \varepsilon > 0$. Suppose $u=0$ is stable.

$\forall \epsilon > 0, \exists \delta_1 = \delta_1(t_0, \epsilon) \text{ s.t.}$

(6)

$$0 \leq u_0 \leq \delta_1 \Rightarrow u(t; t_0, u_0) \leq b(\epsilon) \quad \forall t \geq t_0.$$

Choose $\delta_2 = \delta_2(\epsilon)$ such that $a(\delta_2) < b(\epsilon)$. Define $\delta = \min\{\delta_1, \delta_2\}$

Next, we prove $x=0$ is stable; that is, if $\|x_0\| < \delta \Rightarrow \|x(t)\| < \epsilon, \forall t \geq t_0$

Suppose not, there exist some t^* such that $(\|x_0\| < \delta)$.

$$\|x(t^*)\| \geq \epsilon$$

Let $u_0 = a(\|x_0\|) \leq a(\delta) = a(\delta_2) < \delta_1$ and define $m(t) = V(t, x(t)) - t_0 < t \leq t^*$

Then, by Theorem 12.1, we have

$$V(t, x(t)) \leq \Theta(t; t_0, a(\|x_0\|))$$

by stability of u

$$\Rightarrow b(\epsilon) \leq b(\|x^*\|) \leq V(t^*, x(t^*)) \leq \Theta(t^*, t_0, a(\|x_0\|)) < b(\epsilon)$$

which is a contradiction. This shows $x=0$ is stable. If, moreover, δ is independent of t_0 , then $x=0$ is uniformly stable

Now, we use asymptotic stability of $x=0$ to show as. stability

of $x=0$.

$$\forall b(\epsilon) > 0, \exists \delta_0^* = \delta_0^*(t_0) > 0, T = T(t_0, \epsilon) \text{ such that}$$

$$0 \leq u_0 \leq \delta^* \text{ implies } u(t, t_0, u_0) < b(\epsilon), \forall t \geq t_0 + T.$$

Choose a $\tilde{\delta}$ such that $a(\tilde{\delta}) < \delta_0^*$. $\rho = \min\{\delta_0^*, \tilde{\delta}\}, \|x_0\| < \rho$

$$\Rightarrow b(\|x(t_0)\|) \leq V(t_0, x(t_0)) \leq \Theta(t_0, t_0, a(\|x_0\|)) < b(\epsilon).$$

$$\Rightarrow \|x(t_0)\| \leq b^{-1}(b(\epsilon)). \Rightarrow x=0 \text{ is asymptotically stable}$$

Corollary 12.2. $g(t, u(t)), \beta_k(u_k) = \beta_k u_k.$

(i) In the case $\xi_k = t_k$.

(ii) if $\beta_k > 0$ for any k and the infinite series $\sum_{j=1}^{\infty} \beta_{j-1} (t_j - t_{j-1})$ (18.a)

Converges. then $x=0$ is uniformly stable. ⑦

Proof: (i) (1)
$$\begin{cases} \dot{u}(t) = \beta_k u_{s_k} & , t \in [t_k, t_{k+1}] \\ u(t_0) = u_0 \end{cases}$$

is given by

$$u(t) = \left(1 + \beta_k (t - t_k)\right) \prod_{j=1}^k [1 + \beta_j (t_j - t_{j-1})] u_0$$

By (12.8a), the product $\prod_{j=1}^{\infty} [1 + \beta_j (t_j - t_{j-1})]$ converges

$$\prod_{a=1}^{\infty} (1 + x_a) \leq e^{\sum_{a=1}^{\infty} x_a} < \infty \quad \cdot \quad \prod_{a=1}^{\infty} (1 + x_a) \leq e^{\sum_{a=1}^{\infty} x_a} = e^{\sum_{a=1}^{\infty} \ln(1 + x_a)} \leq e^{\sum_{a=1}^{\infty} x_a} < \infty$$

So that, defining $M = \prod_{j=1}^{\infty} [1 + \beta_j (t_j - t_{j-1})]$

$\Rightarrow u(t, t_0, u_0) = M u_0 < M \delta$, for some δ such that $u_0 < \delta$.

$\Rightarrow u=0$ is uniformly stable.

Corollary 12.3 $g(t, u(t), b_k(u_{s_k})) = \alpha(u(t)) + \beta_k u_{s_k}$

Cor: 12.4 $g(t, u(t), b_k(u_{s_k})) = -w(u) + \beta_k u_{s_k}, w \in K$.

Cor: 12.5 $g(t, u(t), b_k(u_{s_k})) = \alpha(u(t)) + h(t, u(t), b_k(u_{s_k}))$.

Conclusion.

Comparison method + Stability analysis



Stabilize. condition.

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